

Primary Key & Datatypes in SQL query

✓ What is a Primary Key?

A **Primary Key** is a column (or a combination of columns) in a table that:

- **Uniquely identifies each row** in the table.
- **Cannot be NULL.**
- **Must be unique** — no duplicate values are allowed.

Example:

```
1 CREATE TABLE student (  
2     id INT PRIMARY KEY,           -- Primary Key  
3     name VARCHAR(50),  
4     age INT  
5 );  
6
```

💡 In the `student` table, each student has a unique `id`, making it a perfect primary key.

✓ Why Use a Primary Key?

- Ensures **data integrity**.
- Helps relate data across tables (foreign keys).
- Speeds up **searches and joins**.

✚ You can also use `AUTO_INCREMENT` to automatically generate unique IDs:

```
1 CREATE TABLE student (  
2     id INT AUTO_INCREMENT PRIMARY KEY,  
3     name VARCHAR(50),  
4     age INT  
5 );
```

📦 Common SQL Data Types

Here's a table with frequently used SQL data types in MySQL:

Data Type	Description	Example
<code>INT</code>	Integer number	25, 100, -5
<code>VARCHAR(n)</code>	Variable-length text (max <code>n</code> characters)	'John' , 'hello@example.com'
<code>CHAR(n)</code>	Fixed-length text (padded with spaces)	'YES' stored as 'YES '

TEXT	Large blocks of text (more than 255 chars)	Long articles, descriptions
FLOAT	Floating-point number (approx. decimals)	3.14, 99.99
DOUBLE	Double precision floating point	123456.789
DECIMAL(p, s)	Precise decimal number. p = total digits, s = digits after decimal	DECIMAL(5, 2) → 999.99
DATE	Date only (YYYY-MM-DD)	'2024-06-26'
TIME	Time only (HH:MM:SS)	'13:45:00'
DATETIME	Date and time	'2024-06-26 13:45:00'
BOOLEAN / TINYINT(1)	True/false values	1 (true), 0 (false)

💡 **VARCHAR vs TEXT** : Use **VARCHAR** for short fields (name, email); use **TEXT** for long content.

Example:

```

1 CREATE TABLE employee (
2     emp_id INT AUTO_INCREMENT PRIMARY KEY,
3     name VARCHAR(100),
4     salary DECIMAL(10,2),
5     join_date DATE,
6     is_active BOOLEAN
7 );

```

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